



CS39006: NETWORKS LABORATORY

ASSIGNMENT 3: TCP SOCKETS

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PART – 2: WIRESHARK ANALYSIS

1) What are the source and destination IP addresses and ports? Share the screenshots to justify your answer.

*Loopback: lo

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	59130 → 5050 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=0 WS=128
2	0.000022504	127.0.0.1	127.0.0.1	TCP	74	5050 → 59130 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=913456149 WS=128
3	0.000046064	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=913456149 TSecr=913456149
9	6.693238884	127.0.0.1	127.0.0.1	TCP	93	59130 → 5050 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=27 TSval=913462842 TSecr=913456149
10	6.693272110	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=28 Win=65536 Len=0 TSval=913462842 TSecr=913462842
11	6.693558246	127.0.0.1	127.0.0.1	TCP	96	59130 → 5050 [PSH, ACK] Seq=28 Ack=1 Win=65536 Len=30 TSval=913462843 TSecr=913462842
12	6.693575406	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=58 Win=65536 Len=0 TSval=913462843 TSecr=913462843
13	6.693603344	127.0.0.1	127.0.0.1	TCP	67	59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14	6.693611733	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15	6.693978895	127.0.0.1	127.0.0.1	TCP	96	5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843
16	6.694080334	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [ACK] Seq=59 Ack=31 Win=65536 Len=0 TSval=913462843 TSecr=913462843
17	6.694029004	127.0.0.1	127.0.0.1	TCP	67	5050 → 59130 [PSH, ACK] Seq=31 Ack=59 Win=65536 Len=1 TSval=913462843 TSecr=913462843
18	6.694036159	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913462843 TSecr=913462843
19	9.335071080	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [FIN, ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913465484 TSecr=913462843
20	9.375407400	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913465525 TSecr=913465484
21	12.102654285	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [FIN, ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913468252 TSecr=913465484
22	12.102668015	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [ACK] Seq=60 Ack=33 Win=65536 Len=0 TSval=913468252 TSecr=913468252

Source and Destination IPs:

Source IP: 127.0.0.1 (localhost)

Destination IP: 127.0.0.1 (localhost)

Source and Destination Ports:

Source Port: 59130 (client, port dynamically allocated)

Destination Port: 5050 (server)

2) Inspect the Three-way handshaking procedure and capture all packets exchanged for it. Attach the necessary screenshots to demonstrate it.

The Three-Way Handshake in TCP consists of three packets exchanged between a client and a server:

- SYN (Synchronization):** The client sends a SYN packet to the server to initiate the connection.
- SYN-ACK (Synchronization-Acknowledgment):** The server responds with a SYN-ACK packet
- ACK (Acknowledgment):** The client sends back an ACK packet to confirm the connection establishment.

Packet 1: 127.0.0.1:59130 → 127.0.0.1:5050 [SYN] (Client initiates connection)

Packet 2: 127.0.0.1:5050 → 127.0.0.1:59130 [SYN, ACK] (Server responds)

Packet 3: 127.0.0.1:59130 → 127.0.0.1:5050 [ACK] (Client acknowledges)

These three packets complete the TCP connection handshake.

1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	59130 → 5050 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=0 WS=128
2	0.000022504	127.0.0.1	127.0.0.1	TCP	74	5050 → 59130 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=913456149 WS=128
3	0.000046064	127.0.0.1	127.0.0.1	TCP	66	59130 → 5050 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=913456149 TSecr=913456149
9	6.693238884	127.0.0.1	127.0.0.1	TCP	93	59130 → 5050 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=27 TSval=913462842 TSecr=913456149
10	6.693272110	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=28 Win=65536 Len=0 TSval=913462842 TSecr=913462842
11	6.693558246	127.0.0.1	127.0.0.1	TCP	96	59130 → 5050 [PSH, ACK] Seq=28 Ack=1 Win=65536 Len=30 TSval=913462843 TSecr=913462842
12	6.693575406	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=58 Win=65536 Len=0 TSval=913462843 TSecr=913462843
13	6.693603344	127.0.0.1	127.0.0.1	TCP	67	59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14	6.693611733	127.0.0.1	127.0.0.1	TCP	66	5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15	6.693978895	127.0.0.1	127.0.0.1	TCP	96	5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843

▶ Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0
 ▶ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
 ▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
 ▶ Transmission Control Protocol, Src Port: 59130, Dst Port: 5050, Seq: 0, Len: 0

Source Port: 59130
 Destination Port: 5050
 [Stream index: 0]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 1367077082
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 1010 ... = Header Length: 40 bytes (10)
 ▶ **Flags: 0x002 (SYN)**
 000. = Reserved: Not set
 ...0 = Nonce: Not set
0... = Congestion Window Reduced (CWR): Not set
0... = ECN-Echo: Not set
0... = Urgent: Not set
0... = Acknowledgment: Not set
0... = Push: Not set
0... = Reset: Not set
 ▶0...1. = Syn: Set
0...0 = Fin: Not set
 [TCP Flags:S.]
 Window: 65495
 [Calculated window size: 65495]
 Checksum: 0xfe30 [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 ▶ Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
 ▶ [Timestamps]

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1 0.000000000 127.0.0.1 127.0.0.1 TCP 74 59130 → 5050 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=0 WS=128
2 0.000022504 127.0.0.1 127.0.0.1 TCP 74 5050 → 59130 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=913456149 WS=128
3 0.000046064 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=913456149 TSecr=913456149
9 6.693238884 127.0.0.1 127.0.0.1 TCP 93 59130 → 5050 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=27 TSval=913462842 TSecr=913456149
10 6.693272110 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=28 Win=65536 Len=0 TSval=913462842 TSecr=913462842
11 6.693558246 127.0.0.1 127.0.0.1 TCP 96 59130 → 5050 [PSH, ACK] Seq=28 Ack=1 Win=65536 Len=30 TSval=913462843 TSecr=913462842
12 6.693575406 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=58 Win=65536 Len=0 TSval=913462843 TSecr=913462843
13 6.693603344 127.0.0.1 127.0.0.1 TCP 67 59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14 6.693611733 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15 6.693978895 127.0.0.1 127.0.0.1 TCP 96 5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843

* Frame 2: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* Transmission Control Protocol, Src Port: 5050, Dst Port: 59130, Seq: 0, Ack: 1, Len: 0
  Source Port: 5050
  Destination Port: 59130
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 4244562987
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 1367077083
  1010 .... = Header Length: 40 bytes (10)
  * Flags: 0x012 (SYN, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0... = Congestion Window Reduced (CWR): Not set
    ....0... = ECN-Echo: Not set
    ....0... = Urgent: Not set
    ....1... = Acknowledgment: Set
    ....0... = Push: Not set
    ....0... = Reset: Not set
  * ....1... = Syn: Set
    ....0... = Fin: Not set
  [TCP Flags: .....A..S.]
  Window: 65483
  [Calculated window size: 65483]
  Checksum: 0xfe30 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  * Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
  * [Timestamps]
  * [SEQ/ACK analysis]

1 0.000000000 127.0.0.1 127.0.0.1 TCP 74 59130 → 5050 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=0 WS=128
2 0.000022504 127.0.0.1 127.0.0.1 TCP 74 5050 → 59130 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=1 TSval=913456149 TSecr=913456149 WS=128
3 0.000046064 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=913456149 TSecr=913456149
9 6.693238884 127.0.0.1 127.0.0.1 TCP 93 59130 → 5050 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=27 TSval=913462842 TSecr=913456149
10 6.693272110 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=28 Win=65536 Len=0 TSval=913462842 TSecr=913462842
11 6.693558246 127.0.0.1 127.0.0.1 TCP 96 59130 → 5050 [PSH, ACK] Seq=28 Ack=1 Win=65536 Len=30 TSval=913462843 TSecr=913462842
12 6.693575406 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=58 Win=65536 Len=0 TSval=913462843 TSecr=913462843
13 6.693603344 127.0.0.1 127.0.0.1 TCP 67 59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14 6.693611733 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15 6.693978895 127.0.0.1 127.0.0.1 TCP 96 5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843

* Frame 3: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* Transmission Control Protocol, Src Port: 59130, Dst Port: 5050, Seq: 1, Ack: 1, Len: 0
  Source Port: 59130
  Destination Port: 5050
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 1367077083
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 4244562988
  1000 .... = Header Length: 32 bytes (8)
  * Flags: 0x010 (ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0... = Congestion Window Reduced (CWR): Not set
    ....0... = ECN-Echo: Not set
    ....0... = Urgent: Not set
    ....1... = Acknowledgment: Set
    ....0... = Push: Not set
    ....0... = Reset: Not set
    ....0... = Syn: Not set
    ....0... = Fin: Not set
  [TCP Flags: .....A....]
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  * Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  * [Timestamps]
  * [SEQ/ACK analysis]

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3) Inspect the connection closure procedure and capture all packets exchanged for it. Attach the necessary screenshots to demonstrate it.

A TCP connection is typically closed using a Four-Way Handshake, involving the following steps:

- 1) **FIN (Finish) from the client** – The client requests to terminate the connection.
- 2) **ACK from the server** – The server acknowledges the FIN request.
- 3) **FIN from the server** – The server also signals that it is ready to close the connection.
- 4) **ACK from the client** – The client acknowledges the server's FIN, completing the termination.

- a) **Packet 19:** 127.0.0.1:59130 → 127.0.0.1:5050 **[FIN, ACK]** (Client initiates termination)
- b) **Packet 20:** 127.0.0.1:5050 → 127.0.0.1:59130 **[ACK]** (Server acknowledges)
- c) **Packet 21:** 127.0.0.1:59130 → 127.0.0.1:5050 **[FIN, ACK]** (Client initiates termination)
- d) **Packet 22:** 127.0.0.1:5050 → 127.0.0.1:59130 **[ACK]** (Server acknowledges)

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13 6.693603344 127.0.0.1 127.0.0.1 TCP 67 59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14 6.693611733 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15 6.693978895 127.0.0.1 127.0.0.1 TCP 96 5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843
16 6.694090334 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=31 Win=65536 Len=0 TSval=913462843 TSecr=913462843
17 6.694029804 127.0.0.1 127.0.0.1 TCP 67 5050 → 59130 [PSH, ACK] Seq=31 Ack=59 Win=65536 Len=1 TSval=913462843 TSecr=913462843
18 6.694036159 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913462843 TSecr=913462843
19 9.335971680 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [FIN, ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913465484 TSecr=913462843
20 9.375407400 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913465525 TSecr=913465484
21 12.102654285 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [FIN, ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913468252 TSecr=913465484
22 12.102668015 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=60 Ack=33 Win=65536 Len=0 TSval=913468252 TSecr=913468252

> Frame 19: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 59130, Dst Port: 5050, Seq: 59, Ack: 32, Len: 0
  Source Port: 59130
  Destination Port: 5050
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 59 (relative sequence number)
  Sequence Number (raw): 1367077141
  [Next Sequence Number: 60 (relative sequence number)]
  Acknowledgment Number: 32 (relative ack number)
  Acknowledgment number (raw): 4244563019
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x011 (FIN, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0. .... = Congestion Window Reduced (CWR): Not set
    ....0. .... = ECN-Echo: Not set
    ....0. .... = Urgent: Not set
    ....1. .... = Acknowledgment: Set
    ....0. .... = Push: Not set
    ....0. .... = Reset: Not set
    ....0. .... = Syn: Not set
    ....1. .... = Fin: Set
  > [TCP Flags: .....A...F]
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  > [Timestamps]

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13 6.693603344 127.0.0.1 127.0.0.1 TCP 67 59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14 6.693611733 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15 6.693978895 127.0.0.1 127.0.0.1 TCP 96 5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843
16 6.694090334 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=31 Win=65536 Len=0 TSval=913462843 TSecr=913462843
17 6.694029804 127.0.0.1 127.0.0.1 TCP 67 5050 → 59130 [PSH, ACK] Seq=31 Ack=59 Win=65536 Len=1 TSval=913462843 TSecr=913462843
18 6.694036159 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913462843 TSecr=913462843
19 9.335971680 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [FIN, ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913465484 TSecr=913462843
20 9.375407400 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913465525 TSecr=913465484
21 12.102654285 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [FIN, ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913468252 TSecr=913465484
22 12.102668015 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=60 Ack=33 Win=65536 Len=0 TSval=913468252 TSecr=913468252

> Frame 20: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 5050, Dst Port: 59130, Seq: 32, Ack: 60, Len: 0
  Source Port: 5050
  Destination Port: 59130
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 32 (relative sequence number)
  Sequence Number (raw): 4244563019
  [Next Sequence Number: 32 (relative sequence number)]
  Acknowledgment Number: 60 (relative ack number)
  Acknowledgment number (raw): 1367077142
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x010 (ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0. .... = Congestion Window Reduced (CWR): Not set
    ....0. .... = ECN-Echo: Not set
    ....0. .... = Urgent: Not set
    ....1. .... = Acknowledgment: Set
    ....0. .... = Push: Not set
    ....0. .... = Reset: Not set
    ....0. .... = Syn: Not set
    ....0. .... = Fin: Not set
  > [TCP Flags: .....A....]
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  > [Timestamps]
  > [SEQ/ACK analysis]

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```

13 6.693603344 127.0.0.1 127.0.0.1 TCP 67 59130 → 5050 [PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14 6.693611733 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15 6.693978895 127.0.0.1 127.0.0.1 TCP 96 5050 → 59130 [PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843
16 6.694090334 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=31 Win=65536 Len=0 TSval=913462843 TSecr=913462843
17 6.694029804 127.0.0.1 127.0.0.1 TCP 67 5050 → 59130 [PSH, ACK] Seq=31 Ack=59 Win=65536 Len=1 TSval=913462843 TSecr=913462843
18 6.694036159 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913462843 TSecr=913462843
19 9.335971680 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [FIN, ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913465484 TSecr=913462843
20 9.375407400 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913465525 TSecr=913465484
21 12.102654285 127.0.0.1 127.0.0.1 TCP 66 5050 → 59130 [FIN, ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913468252 TSecr=913465484
22 12.102668015 127.0.0.1 127.0.0.1 TCP 66 59130 → 5050 [ACK] Seq=60 Ack=33 Win=65536 Len=0 TSval=913468252 TSecr=913468252

> Frame 21: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 5050, Dst Port: 59130, Seq: 32, Ack: 60, Len: 0
  Source Port: 5050
  Destination Port: 59130
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 32 (relative sequence number)
  Sequence Number (raw): 4244563019
  [Next Sequence Number: 33 (relative sequence number)]
  Acknowledgment Number: 60 (relative ack number)
  Acknowledgment number (raw): 1367077142
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x011 (FIN, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Nonce: Not set
    ....0. .... = Congestion Window Reduced (CWR): Not set
    ....0. .... = ECN-Echo: Not set
    ....0. .... = Urgent: Not set
    ....1. .... = Acknowledgment: Set
    ....0. .... = Push: Not set
    ....0. .... = Reset: Not set
    ....0. .... = Syn: Not set
    ....1. .... = Fin: Set
  > [TCP Flags: .....A...F]
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  > [Timestamps]

```


13	6.693603344	127.0.0.1	127.0.0.1	TCP	67	59130	→	5050	[PSH, ACK] Seq=58 Ack=1 Win=65536 Len=1 TSval=913462843 TSecr=913462843
14	6.693611733	127.0.0.1	127.0.0.1	TCP	66	5050	→	59130	[ACK] Seq=1 Ack=59 Win=65536 Len=0 TSval=913462843 TSecr=913462843
15	6.693578895	127.0.0.1	127.0.0.1	TCP	96	5050	→	59130	[PSH, ACK] Seq=1 Ack=59 Win=65536 Len=30 TSval=913462843 TSecr=913462843
16	6.694090334	127.0.0.1	127.0.0.1	TCP	66	59130	→	5050	[ACK] Seq=59 Ack=31 Win=65536 Len=0 TSval=913462843 TSecr=913462843
17	6.694029804	127.0.0.1	127.0.0.1	TCP	67	5050	→	59130	[PSH, ACK] Seq=31 Ack=59 Win=65536 Len=1 TSval=913462843 TSecr=913462843
18	6.694036159	127.0.0.1	127.0.0.1	TCP	66	59130	→	5050	[ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913462843 TSecr=913462843
19	9.335071680	127.0.0.1	127.0.0.1	TCP	66	59130	→	5050	[FIN, ACK] Seq=59 Ack=32 Win=65536 Len=0 TSval=913465484 TSecr=913462843
20	9.375407400	127.0.0.1	127.0.0.1	TCP	66	5050	→	59130	[ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913465525 TSecr=913465484
21	12.102654285	127.0.0.1	127.0.0.1	TCP	66	5050	→	59130	[FIN, ACK] Seq=32 Ack=60 Win=65536 Len=0 TSval=913468252 TSecr=913465484
22	12.102660915	127.0.0.1	127.0.0.1	TCP	66	59130	→	5050	[ACK] Seq=60 Ack=33 Win=65536 Len=0 TSval=913468252 TSecr=913468252

* Frame 22: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
 * Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
 * Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
 * Transmission Control Protocol, Src Port: 59130, Dst Port: 5050, Seq: 60, Ack: 33, Len: 0
 Source Port: 59130
 Destination Port: 5050
 [Stream index: 0]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 0]
 Sequence Number: 60 (relative sequence number)
 Sequence Number (raw): 1367077142
 [Next Sequence Number: 60 (relative sequence number)]
 Acknowledgment Number: 33 (relative ack number)
 Acknowledgment number (raw): 4244563020
 1000 ... = Header Length: 32 bytes (0)
 * Flags: 0x010 (ACK)
 0000. = Reserved: Not set
 ...0. = Nonce: Not set
0. = Congestion Window Reduced (CWR): Not set
0. = ECN-Echo: Not set
0. = Urgent: Not set
0. = Acknowledgment: Set
0. = Push: Not set
0. = Reset: Not set
0. = Syn: Not set
0. = Fin: Not set
 [TCP Flags:A....]
 Window: 512
 [Calculated window size: 65536]
 [Window size scaling factor: 128]
 Checksum: 0x140b [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 * Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 * [Timestamps]
 * [SEQ/ACK analysis]

4) Inspect the traffics and count the number of packets exchanged for the transfer of a file(related to data only) between client and server. Plot a graph 'file size vs the number of packets' clearly based on your observation.

NOTE: I calculated the total packet exchanged and time taken by modifying the client and server code to make these calculation.

The program calculates the number of packets transferred while receiving the encrypted file by maintaining a counter n in the write_file() and write_encrypted_file() function. Each time a chunk of data is received using recv(), the counter is incremented. The loop continues until the terminating character \$ is detected, indicating the end of the file. The total number of packets received is then printed. The sum of packets exchanged between client and server is the number of packets exchanged for the transfer of a file(related to data only) (ACK not counted).

The total encryption time is measured using clock_t in the main() function. The time before sending the file is recorded (t1), and the time after receiving the encrypted file is recorded (t2). The difference (t2 - t1) is converted into milliseconds using CLOCKS_PER_SEC, and the total time taken for encryption is displayed.

```

tuhin@tuhin-hp15s-fy5xxx:~/Computer Networks/LA3/Server$ ./server
[+] Server socket created.
[+] Binded to port 5050
[+] Listening...
[+] Connection accepted from 127.0.0.1:57148.
[127.0.0.1 : 57148][+] 11 packets transferred successfully.
[127.0.0.1 : 57148][+] File encrypted successfully.
[127.0.0.1 : 57148][+] Encrypted file sent successfully.

[127.0.0.1 : 57148][+] 103 packets transferred successfully.
[127.0.0.1 : 57148][+] File encrypted successfully.
[127.0.0.1 : 57148][+] Encrypted file sent successfully.

[127.0.0.1 : 57148][+] 1025 packets transferred successfully.
[127.0.0.1 : 57148][+] File encrypted successfully.
[127.0.0.1 : 57148][+] Encrypted file sent successfully.

[127.0.0.1 : 57148][+] 10486 packets transferred successfully.
[127.0.0.1 : 57148][+] File encrypted successfully.
[127.0.0.1 : 57148][+] Encrypted file sent successfully.
  
```

```

● tuhin@tuhi-hp15s-fy5xxx:~/Computer Networks/LA3/Client$ ./client
[+] Client Socket is created.
[+] Connected to Server.
[*] Enter file name: file_1KB.txt
[*] Enter encryption key: QWERTYUIOPASDFGHJKLZXCVBNM

[19:06:26] Encryption key sent successfully.
[19:06:26] File sent successfully.
[19:06:26] Received 12 packets from server.
[19:06:26] Encrypted File received successfully.

[+] Received file can be found in file_1KB.txt.enc
[+] Total time taken to encrypt file: 0.476000 ms

[*] Do you want to send another file? (Yes/No): Yes
[*] Enter file name: file_10KB.txt
[*] Enter encryption key: QWERTYUIOPASDFGHJKLZXCVBNM

[19:06:54] Encryption key sent successfully.
[19:06:54] File sent successfully.
[19:06:54] Received 103 packets from server.
[19:06:54] Encrypted File received successfully.

[+] Received file can be found in file_10KB.txt.enc
[+] Total time taken to encrypt file: 1.677000 ms

[*] Do you want to send another file? (Yes/No): Yes
[*] Enter file name: file_100KB.txt
[*] Enter encryption key: QWERTYUIOPASDFGHJKLZXCVBNM

[19:07:44] Encryption key sent successfully.
[19:07:44] File sent successfully.
[19:07:44] Received 1025 packets from server.
[19:07:44] Encrypted File received successfully.

[+] Received file can be found in file_100KB.txt.enc
[+] Total time taken to encrypt file: 5.759000 ms

[*] Do you want to send another file? (Yes/No): Yes
[*] Enter file name: file_1MB.txt
[*] Enter encryption key: QWERTYUIOPASDFGHJKLZXCVBNM

[19:08:04] Encryption key sent successfully.
[19:08:04] File sent successfully.
[19:08:04] Received 10487 packets from server.
[19:08:04] Encrypted File received successfully.

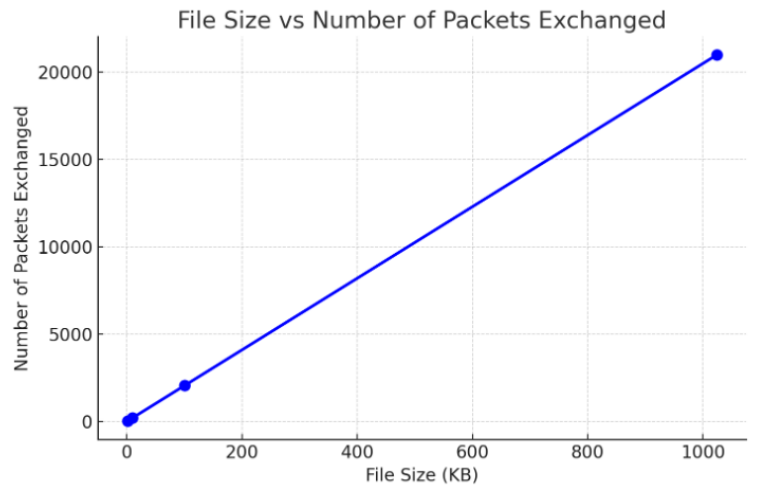
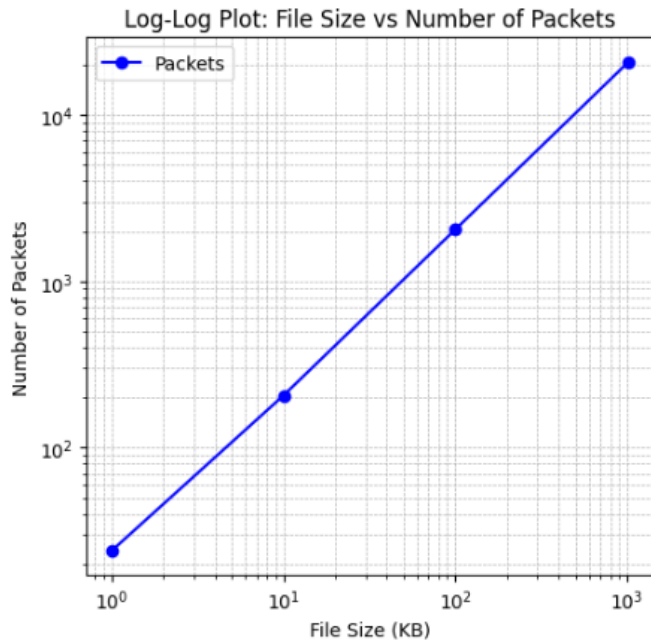
[+] Received file can be found in file_1MB.txt.enc
[+] Total time taken to encrypt file: 27.086000 ms

[*] Do you want to send another file? (Yes/No): No
[+] Client socket disconnected from server

```

File Size	Packets Exchanged	Time Taken (ms)
1KB	23	0.476
10KB	206	1.677
100KB	2050	5.759
1MB	20974	27.086

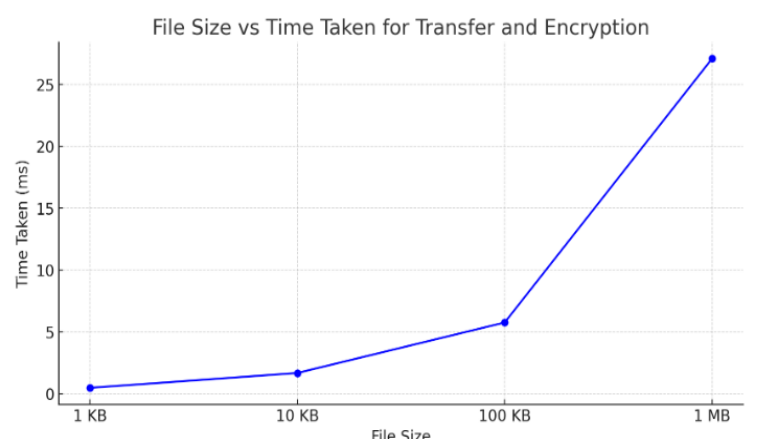
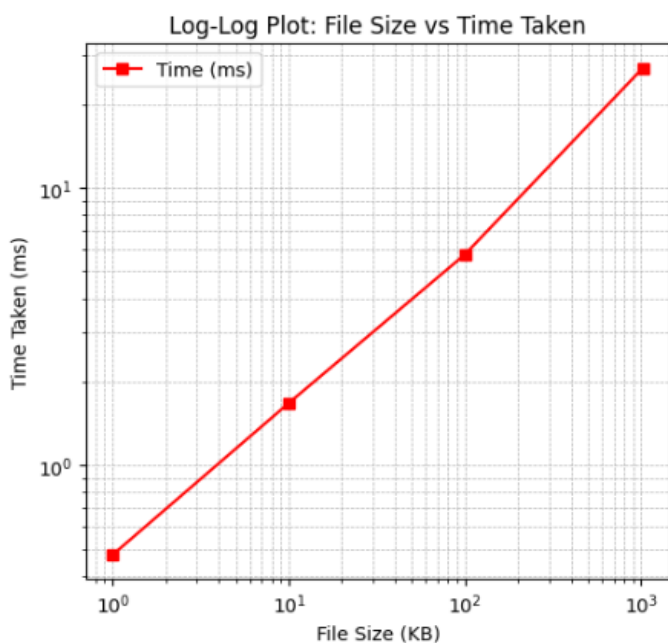
Trend Analysis: As the file size increases, the number of packets exchanged between the client and server also increases significantly. The relationship between file size and the number of packets appears to be linear, as indicated by a proportional increase in packets as file size grows.



5) Measure the total time taken for the file transfer , its encryption and send back it from server to the client. Plot a graph 'file size vs time' clearly based on your observation and also attach the necessary screenshots.

File Size	Packets Exchanged	Time Taken (ms)
1KB	23	0.476
10KB	206	1.677
100KB	2050	5.759
1MB	20974	27.086

Trend Analysis: The time taken grows modestly from 0.476 ms to 1.677 ms, indicating that the communication overhead dominates for smaller file sizes. The time taken increases more sharply, reaching 5.759 ms. The larger number of packets exchanged becomes more significant, suggesting that transfer time is starting to impact performance. A noticeable jump in time is observed from 5.759 ms to 27.086 ms. The transfer and encryption overhead dominate due to the exponential increase in the number of packets (from 2050 to 20974).



6) Calculate the average size packet exchanged during the data communication? Take reference of the plotted graph in the above question.

Since the file is exchanged back and forth between the client and server, each file size effectively doubles for the communication process. To calculate the average packet size for data communication, we can use the formula:

Average Packet Size = $2 \times \text{File Size in Bytes} / \text{Number of Packets Exchanged}$

File Size (Bytes)	Packets Exchanged	Average Packet Size (Bytes)
1024	23	$2 \times 1024 / 23 \approx 89.04$
10240	206	$2 \times 10240 / 206 \approx 99.61$
102400	2050	$2 \times 102400 / 2050 \approx 99.90$
1048576	20974	$2 \times 1048576 / 20974 \approx 99.95$

Total File Size = $1024 + 10240 + 102400 + 1048576 = 1162240$ Bytes

Since the file is exchanged back and forth, we double the total file size: $2 \times 1162240 = 2324480$ Bytes

Total Number of Packets = $23 + 206 + 2050 + 20974 = 23353$ Packets

Overall Average Packet Size = $2324480 / 23353 \approx 99.52$ Bytes

The overall average packet size is **approximately 99.52 Bytes**.